Preconditioned Primal-Dual Gradient Method for Nonconvex Nonlinear Composite Optimization

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1 Algorithm

x 更新时约束 $g(x)$ 在 x^k 处线性化，y 更新时约束 $g(x)$ 在 x^{k+1} 处线性化，在原始变量上做外推,

$$
\begin{aligned}
x^{k+1} &= \arg\min_x \left\{ \langle \nabla f(x^k), x - x^k \rangle + \langle y^k, g(x^k) + J(x^k)(x - x^k) \rangle + \frac{1}{2\tau} \|x - x^k\|^2 \right\} \\
&= \arg\min_x \left\{ \langle \nabla f(x^k) + J(x^k)^T y^k, x - x^k \rangle + \frac{1}{2\tau} \|x - x^k\|^2 \right\} \\
\tilde{x}^{k+1} &= 2x^{k+1} - x^k \\
y^{k+1} &= \arg\min_y \left\{ h^*(y) - \langle g(x^{k+1}) + J(x^{k+1})(\tilde{x}^{k+1} - x^{k+1}), y \rangle + \frac{1}{2} \|y - y^k\|_{M_k}^2 \right\} \\
&= \arg\min_y \left\{ h^*(y) - \langle g^{k+1}(\tilde{x}^{k+1})), y \rangle + \frac{1}{2} \|y - y^k\|_{M_k}^2 \right\}
\end{aligned}
$$

其中，$M_k = \tau J(x^k) J(x^k)^T$

1.1 Preliminaries

Notations

$\|A\| = \|A\|_F$, $\|a\| = \|a\|_2$

一些有用的性质

1. 无穷连续可微函数：$\forall i = 1, 2, \cdots, m$，$g_i(x)$ $L_{g_i}^0$-Lipschitz 连续: $\|\nabla g_i(x)\| \leq L_{g_i}^0$;
   $\nabla g_i(x)$ $L_{g_i} - Lipschitz$ 连续：$H_{g_i}(x) \preceq L_{g_i} I$，即 $\lambda_{max}(H_{g_i}(x)) \leq L_{g_i}, \forall x \in \mathcal{X}$

2. 非线性映射 $g(x) : \mathbb{R}^n \to \mathbb{R}^m = (g_1(x), \cdots, g_m(x))^T$ $L_g^0 - Lipschitz$ 连续，$L_g^0 := \sum_{i=1}^m L_{g_i}$

$$\|g(x) - g(x')\| \leq L_g^0 \|x - x'\|$$

且 Jacobian $J(x)$ 有界：$\forall x \in \mathcal{X}$

$$\|J(x)\| = \sqrt{\sum_{i=1}^{m} \|\nabla g_i(x)\|^2} \leq \sqrt{\sum_{i=1}^{m} (L_{g_i}^0)^2} \leq \sum_{i=1}^{m} L_{g_i}^0 = L_g^0 \qquad (\|A\|_F \text{ 定义，和范数关系})$$

以下分析使用更 tight 的上界，即 $\|J(x)\| \leq \sqrt{\sum_{i=1}^{m} (L_{g_i}^0)^2} := M_g^0$

算法更新的最优性条件

1. x 更新的最优性条件：

$$x^{k+1} = x^k - \tau \left(\nabla f(x^k) + (\nabla J(x^k))^T y^k \right)$$

2. 在 x 上做外推：

$$\tilde{x}^{k+1} = 2x^{k+1} - x^k, \quad \tilde{x}^k = 2x^k - x^{k-1}$$

3. y 更新的最优性条件：

$$g^{k+1} = -M_k \left(y^{k+1} - y^k \right) + g(\tilde{x}^{k+1}), \quad g^{k+1} \in \partial h^*(y^{k+1})$$

2 收敛性分析

Lemma 1 (dual boundedness)

Proof

$$\begin{aligned}
\frac{\delta}{2}\|J(x^k)^T(y^{k+1} - y^k)\|^2 &= \frac{\delta}{2}\|(J(x^k) - J(x^{k+1}))^T y^{k+1} + J(x^{k+1})^T y^{k+1} - J(x^k)^T y^k\|^2 \\
&\leq \delta\|(y^{k+1})^T(J(x^{k+1}) - J(x^k))\|^2 + \delta\|J(x^{k+1})^T y^{k+1} - J(x^k)^T y^k\|^2 \\
&\leq \delta \sum_{i=1}^{m} \|y^{k+1}\|^2 \|\nabla g_i(x^{k+1}) - \nabla g_i(x^k)\|^2 \\
&\quad + \delta\| - \nabla f(x^{k+1}) + \frac{x^{k+1} - x^{k+2}}{\tau} + \nabla f(x^k) + \frac{x^{k+1} - x^k}{\tau}\|^2 \\
&\qquad\qquad (x^{k+2} - x^{k+1} = -\tau(\nabla f(x^{k+1}) + J(x^{k+1})^T y^{k+1})) \\
&\leq \delta M^2 \left(\sum_{i=1}^{m} L_{g_i}^2 \right) \|x^{k+1} - x^k\|^2 \\
&\quad + 2\delta(\frac{1}{\tau} + L_f)^2 \|x^{k+1} - x^k\|^2 + \frac{2\delta}{\tau^2}\|x^{k+2} - x^{k+1}\|^2 \\
&\leq (\delta M^2 \left(\sum_{i=1}^{m} L_{g_i}^2 \right) + 2\delta(\frac{1}{\tau} + L_f)^2)\|x^{k+1} - x^k\|^2 + \frac{2\delta}{\tau^2}\|x^{k+2} - x^{k+1}\|^2 \qquad (2.1)
\end{aligned}$$

Lemma 2 (sufficient descent)

Proof 首先，根据在原始变量做外推的算法更新的迭代公式和 h^* 的凸性，

$$\begin{aligned} L(x^{k+1},y^{k+1})-L(x^k,y^k) \le & -(\frac{1}{\tau}-\frac{L_f}{2})\|x^{k+1}-x^k\|^2 \\ & +\langle y^{k+1},g(x^{k+1})\rangle-\langle y^k,g^k(x^{k+1})\rangle+\langle g^k,y^k-y^{k+1}\rangle \end{aligned} \tag{2.2}$$

对于(2.2)右侧的第二项，为了和最后一项合并，凑 $y^{k+1}-y^k$，(2.2)等价变形成如下形式：

$$\begin{aligned} L(x^{k+1},y^{k+1})-L(x^k,y^k) \le & -(\frac{1}{\tau}-\frac{L}{2})\|x^{k+1}-x^k\|^2 \\ & +\langle y^{k+1}-y^k,g(x^{k+1})-g^k\rangle+\langle y^k,g(x^{k+1})-g^k(x^{k+1})\rangle \end{aligned} \tag{2.3}$$

对于 (2.3) 右侧的第二项，根据 y^k 更新的最优性条件：$0\in\partial h^*(y^k)-g^k(\tilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})$ 且取 $g^k\in\partial h^*(y^k)$，

$$\begin{aligned} \langle y^{k+1}-y^k,g(x^{k+1})-g^k\rangle &= \langle y^{k+1}-y^k,g(x^{k+1})-g^k(\tilde{x}^k)\rangle+\frac{1}{\sigma}\langle y^{k+1}-y^k,y^k-y^{k-1}\rangle \\ &= \langle y^{k+1}-y^k,g(x^{k+1})-g^k(x^{k+1})\rangle \\ &\quad +\langle y^{k+1}-y^k,g^k(x^{k+1})-g^k(\tilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\rangle \end{aligned} \tag{2.4}$$

将 (2.4) 带入 (2.3)有，

$$\begin{aligned} L(x^{k+1},y^{k+1})-L(x^k,y^k) \le & -(\frac{1}{\tau}-\frac{L}{2})\|x^{k+1}-x^k\|^2 \\ & +\langle y^{k+1},g(x^{k+1})-g^k(x^{k+1})\rangle \\ & +\langle y^{k+1}-y^k,g^k(x^{k+1})-g^k(\tilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\rangle \end{aligned} \tag{2.5}$$

以下，讨论(2.5)的放缩。

对 (2.5) 右侧的第二项进行放缩，

$$\begin{aligned} \langle y^{k+1},g(x^{k+1})-g^k(x^{k+1})\rangle &\le \|y^{k+1}\|\|g(x^{k+1})-g^k(x^{k+1})\| \\ &\le M\|g(x^{k+1})-g^k(x^{k+1})\| \quad (\|y^k\|\le M) \\ &\le M\sum_{i=1}^{m}\|g_i(x^{k+1})-g_i^k(x^{k+1})\| \\ &\le \frac{ML_g^0}{2}\|x^k-x^{k+1}\|^2 \end{aligned} \tag{2.6}$$

利用迭代公式对(2.5)右侧最后一项进行等价变形，

$$\begin{aligned}
\langle y^{k+1}-y^k, g^k(x^{k+1})-g^k(\tilde{x}^k)+\frac{1}{\sigma}(y^k-y^{k-1})\rangle &= \langle y^{k+1}-y^k, J(x^k)[(x^{k+1}-x^k)-(x^k-x^{k-1})]+\frac{1}{\sigma}(y^k-y^{k-1})\rangle \\
&\qquad (g^k(z)=g(x^k)+J(x^k)(z-x^k),\ \tilde{x}^k=2x^k-x^{k-1}) \\
&= \langle y^{k+1}-y^k, \tau J(x^k)(\nabla f(x^{k-1})-\nabla f(x^k)+J(x^{k-1})^T y^{k-1}-J(x^k)^T y^k)+\frac{1}{\sigma}(y^k-y^{k-1})\rangle \\
&\qquad (x^k-x^{k-1}=-\tau(\nabla f(x^{k-1})+J(x^{k-1})^T y^{k-1})) \\
&= \langle J(x^k)^T(y^{k+1}-y^k), \nabla f(x^{k-1})-\nabla f(x^k)\rangle \\
&\quad + \langle y^{k+1}-y^k, \left(\frac{1}{\sigma}I-\tau J(x^k)J(x^k)^T\right)(y^k-y^{k-1})\rangle + \langle y^{k+1}-y^k, \tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\rangle
\end{aligned} \tag{2.7}$$

对于(2.7)中的 $\langle y^{k+1}-y^k, \left(\frac{1}{\sigma}I-\tau J(x^k)J(x^k)^T\right)(y^k-y^{k-1})\rangle$，如果此时令 $\frac{1}{\sigma}I=\frac{1}{\sigma^k}I=M_k$，那么这项就没了! 然后继续放缩(2.7)，因为不太想出现 y^k 和 y^{k+1}，我们再次合并(2.7)中的剩余两项，

$$\begin{aligned}
&\langle y^{k+1}-y^k, g^k(x^{k+1})-g^k(\tilde{x}^k)+M_k(y^k-y^{k-1})\rangle \\
&= \langle J(x^k)^T(y^{k+1}-y^k), \nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\rangle \\
&\le \frac{\delta}{2}\|J(x^k)^T(y^{k+1}-y^k)\|^2+\frac{1}{2\delta}\|\nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\|^2 \\
&\qquad (\langle a,b\rangle \le \frac{\delta}{2}\|a\|^2+\frac{1}{2\delta}\|b\|^2)
\end{aligned} \tag{2.8}$$

对于(2.8)中的第二项放缩，

$$\begin{aligned}
&\frac{1}{2\delta}\|\nabla f(x^{k-1})-\nabla f(x^k)+\tau J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\|^2 \qquad (2.9)\\
&\quad \le \frac{1}{\delta}\|\nabla f(x^k)-\nabla f(x^{k-1})\|^2+\frac{\tau^2}{\delta}\|J(x^k)\left(J(x^{k-1})-J(x^k)\right)^T y^{k-1}\|^2 \\
&\quad \le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\|\sum_{i=1}^m (y^{k-1})^T(J(x^{k-1})-J(x^k))(J(x^k))^T\|^2 \\
&\quad \le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\sum_{i=1}^m \|(y^{k-1})^T(\nabla g_i(x^{k-1})-\nabla g_i(x^k))(\nabla g_i(x^k))^T\|^2 \\
&\quad \le \frac{L_f^2}{\delta}\|x^k-x^{k-1}\|^2+\frac{\tau^2}{\delta}\sum_{i=1}^m \|y^{k-1}\|^2\|\nabla g_i(x^k)\|^2\|\nabla g_i(x^{k-1})-\nabla g_i(x^k)\|^2 \\
&\quad \le \frac{L_f^2+\sum_{i=1}^m(\tau M L_{g_i}^0 L_{g_i})^2}{\delta}\|x^k-x^{k-1}\|^2 \qquad (2.10)
\end{aligned}$$

对于(2.8)中的第一项放缩，

$$\begin{aligned}
\frac{\delta}{2}\|J(x^k)^T(y^{k+1}-y^k)\|^2 &= \frac{\delta}{2}\|(J(x^k)-J(x^{k+1}))^T y^{k+1}+J(x^{k+1})^T y^{k+1}-J(x^k)^T y^k\|^2 \\
&\le \delta\|(y^{k+1})^T(J(x^{k+1})-J(x^k))\|^2+\delta\|J(x^{k+1})^T y^{k+1}-J(x^k)^T y^k\|^2 \\
&\le \delta\sum_{i=1}^m \|y^{k+1}\|^2\|\nabla g_i(x^{k+1})-\nabla g_i(x^k)\|^2 \\
&\quad +\delta\|-\nabla f(x^{k+1})+\frac{x^{k+1}-x^{k+2}}{\tau}+\nabla f(x^k)+\frac{x^{k+1}-x^k}{\tau}\|^2 \\
&\qquad\qquad (x^{k+2}-x^{k+1}=-\tau(\nabla f(x^{k+1})+J(x^{k+1})^T y^{k+1})) \\
&\le \delta M^2\left(\sum_{i=1}^m L_{g_i}^2\right)\|x^{k+1}-x^k\|^2 \\
&\quad +2\delta(\frac{1}{\tau}+L_f)^2\|x^{k+1}-x^k\|^2+\frac{2\delta}{\tau^2}\|x^{k+2}-x^{k+1}\|^2 \\
&\le (\delta M^2\left(\sum_{i=1}^m L_{g_i}^2\right)+2\delta(\frac{1}{\tau}+L_f)^2)\|x^{k+1}-x^k\|^2+\frac{2\delta}{\tau^2}\|x^{k+2}-x^{k+1}\|^2 \qquad (2.11)
\end{aligned}$$

综合上述不等式，可以得到

$$\begin{aligned}
L(x^{k+1},y^{k+1})-L(x^k,y^k) \le -(\frac{1}{\tau}-\frac{L}{2})\|x^{k+1}-x^k\|^2 &+\langle y^{k+1}-y^k, g(x^{k+1})-g(\tilde{x}^k)\rangle \\
&+\frac{1}{\sigma}\langle y^{k+1}-y^k, y^k-y^{k-1}\rangle \\
&+\langle y^k, g(x^{k+1})-g^k(x^{k+1})\rangle \\
&\quad (h^*\text{的凸性, } y^k\text{的更新公式, } x^{k+1}\text{的更新公式})
\end{aligned}$$

其中，$g(\tilde{x}^k)=g(2x^k-x^{k-1})$，$g^k(x^{k+1})=g(x^k)+J(x^k)(x^{k+1}-x^k)$ 不等号左侧第二项放缩：

$$\begin{aligned}
\langle y^{k+1}-y^k, g(x^{k+1})-g(\tilde{x}^k)\rangle &\le \frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{1}{2\delta}\|g(x^{k+1})-g(\tilde{x}^k)\|^2 \\
&\le \frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{L}{2\delta}\|x^{k+1}-x^k+x^{k-1}-x^k\|^2 \\
&\qquad (g_i(u)\text{是}L_{g_i}\text{ L-连续的}) \\
&\le \frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{L\tau^2}{2\delta}\|\nabla f(x^{k-1})-\nabla f(x^k)+J(x^{k-1})^T y^{k-1}-J(x^k)^T y^k\|^2 \\
&\qquad (x^{k+1}=x^k-\tau(\nabla f(x^k)+J(x^k)^T y^k)), \text{且用了 } x^k \text{ 的更新}) \\
&\le \frac{\delta}{2}\|y^{k+1}-y^k\|^2+\frac{LL_f^2\tau^2}{\delta}\|x^{k-1}-x^k\|^2+\frac{L\tau^2}{\delta}\|J(x^{k-1})^T y^{k-1}-J(x^k)^T y^k\|^2 \\
&\qquad (f(x)\text{是}L_f\text{-smooth 的})
\end{aligned}$$

在想标红的这部分，可不可以对 Jacobian 做一个类似于线性（Ax）的假设，使得这部分可以放缩成 $c\|y^k-y^{k-1}\|^2$；

不等号左侧第四项放缩: 结合上午的讨论和之前讲的弱凸文章里的 one-side accuracy 假设:

$$\begin{aligned}\langle y^k, g(x^{k+1}) - g^k(x^{k+1})\rangle &\le M\|g(x^{k+1}) - g^k(x^{k+1})\| \quad (\|y^k\| \le M)\\ &\le \frac{M\mu}{2}\|x^k - x^{k+1}\|^2\\ &(g_i(u)\text{无穷连续可微，且在序列有界的假设下 L-smooth，海瑟阵可以被 bounded})\end{aligned}$$

不等式第三项放缩：没办法像线性约束那样消掉，考虑如下放缩

$$\frac{1}{\sigma}\langle y^{k+1} - y^k, y^k - y^{k-1}\rangle \le \frac{1}{2\sigma}(\|y^k - y^{k+1}\|^2 + \|y^k - y^{k-1}\|^2)$$

$$L(x, y) = f(x) + \langle y, g(x)\rangle - h^*(y) \tag{2.12}$$

Lemma 3 (subgradient boundedness) 定义 $\kappa_1 = \frac{14}{\tau^2}$, $\kappa_2 = 3\left(L_f^2 + \frac{1}{\tau^2} + 4M^2\sum_{i=1}^m L_{g_i}^2 + 4\left(\frac{1}{\tau} + L_f\right)^2\right)$,
$\kappa_3 = 2(M_g^0)^2\left(1 + \tau^2\left(M^2\sum_{i=1}^m L_{g_i}^2 + 2\left(\frac{1}{\tau} + L_f\right)^2\right)\right)$,
$\kappa = \max\{\kappa_1, \kappa_2 + \kappa_3\}$, $u^k = \{x^k, x^{k-1}\}$, 于是有,

$$\begin{aligned}\operatorname{dist}(0, \partial L(x^k, y^k)) &\le \kappa_1\|x^{k+1} - x^k\|^2 + (\kappa_2 + \kappa_3)\|x^k - x^{k-1}\|^2 &(2.13)\\ &\le \kappa\|u^{k+1} - u^k\|^2 &(2.14)\end{aligned}$$

$0 \in \partial L(x^k, y^k)$

$$\begin{aligned}d_x^k &= \nabla_x L(x^k) + J(x^k)^T y^k\\ \partial L(x^k) \ni d_y^k &= g(x^k) - g_y^k, \text{ where } g^k \in \partial h^*(y^k)\end{aligned}$$

第一部分：根据 x^k 更新的最优性条件,

$$\begin{aligned}\|d_1^k\|^2 &= \|(\nabla f(x^k) - \nabla f(x^{k-1})) + (\nabla f(x^{k-1}) + J(x^{k-1})^T y^{k-1}) + (J(x^k)^T y^k - J(x^{k-1})^T y^{k-1})\|^2 &(2.15)\\ &\le 3(\|\nabla f(x^k) - \nabla f(x^{k-1})\|^2 + \|\nabla f(x^{k-1}) + J(x^{k-1})^T y^{k-1}\|^2 + \|J(x^k)^T y^k - J(x^{k-1})^T y^{k-1}\|^2)\\ &\le 3(L_f^2 + \frac{1}{\tau^2})\|x^k - x^{k-1}\|^2 + 3\|J(x^k)^T y^k - J(x^{k-1})^T y^k + J(x^{k-1})^T y^k - J(x^{k-1})^T y^{k-1}\|^2\\ &\le 3(L_f^2 + \frac{1}{\tau^2})\|x^k - x^{k-1}\|^2 + 6\|(J(x^k) - J(x^{k-1}))^T y^k\|^2 + 6\|J(x^{k-1})^T(y^k - y^{k-1})\|^2\\ &\le 3(L_f^2 + \frac{1}{\tau^2} + 2M^2\left(\sum_{i=1}^m L_{g_i}^2\right))\|x^k - x^{k-1}\|^2 &(2.16)\\ &+ 6(M^2\left(\sum_{i=1}^m L_{g_i}^2\right) + 2(\frac{1}{\tau} + L_f)^2)\|x^k - x^{k-1}\|^2 + \frac{12}{\tau^2}\|x^{k+1} - x^k\|^2\\ &= \frac{12}{\tau^2}\|x^{k+1} - x^k\|^2 + 3(L_f^2 + \frac{1}{\tau^2} + 4M^2\left(\sum_{i=1}^m L_{g_i}^2\right) + 4(\frac{1}{\tau} + L_f)^2)\|x^k - x^{k-1}\|^2\end{aligned}$$

其中，不等式(2.16)是根据 Lemma 1、$J(x^k)$ 的 Lipschiz 连续性及范数相容性得到。第二部分：根据 y